

Patient Information

Gender: Female
Age: 22
Hypertension: 0
Heart Disease: 1
Smoking History: current
Bmi: 25.25
Hba1C Level: 5.76
Blood Glucose Level: 128
Bmi Category: Overweight
Age Group: 21-40

Prediction

Prediction: Low Diabetes Risk
Risk Probability: 4.00%

AI Clinical Summary

Clinical Summary

The Random Forest model predicts a **Low Diabetes Risk** with a probability of 4.00% for this patient. However, it is crucial to note that while the model indicates low risk, the patient's individual clinical markers warrant close attention. Specifically, her HbA1c level of 5.76% and Blood Glucose Level of 128 mg/dL fall into the prediabetes to diabetes range according to standard clinical guidelines. This discrepancy highlights the importance of integrating AI predictions with comprehensive clinical judgment and patient context.

Major Risk Factors

Based on the provided patient information, the following factors are clinically relevant and typically contribute to diabetes risk:

- * **Elevated HbA1c Level:** At 5.76%, this is in the prediabetes range (5.7-6.4%).
- * **Elevated Blood Glucose Level:** At 128 mg/dL, this falls into the diabetes range (≥ 126 mg/dL) or high prediabetes range.
- * **Current Smoking History:** Smoking is a known risk factor for type 2 diabetes and cardiovascular disease.
- * **History of Heart Disease:** This is a significant comorbidity often linked with metabolic issues.
- * **BMI:** At 25.25 kg/m², this is at the upper end of the normal weight range, bordering overweight.

Lifestyle Recommendations

- Healthy Eating:** Focus on a balanced diet rich in whole grains, lean proteins, fruits, and vegetables, while limiting processed foods, sugary drinks, and unhealthy fats.
- Regular Physical Activity:** Aim for at least 150 minutes of moderate-intensity aerobic exercise per week, combined with strength training.
- Smoking Cessation:** Strongly encourage and support efforts to quit smoking to improve overall health and reduce various disease risks.
- Weight Management:** Work towards maintaining a healthy weight through consistent healthy eating and regular exercise.

Important Note

This summary is an AI-generated educational interpretation of a machine learning prediction and is not a medical diagnosis or a substitute for professional medical advice. All healthcare decisions should be made by a qualified clinician in consultation with the patient.

Disclaimer: *This report is generated for educational purposes only and is not a medical diagnosis.*